Paper 6

AI-Assisted Portfolio Triage for Canada's Major Projects: Deploying a Risk Engine Dashboard

Running Title: AI Portfolio Triage – Dashboard Deployment

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Abstract

Canada's Major Projects Inventory (MPI) records more than 500 large-scale energy, mining, and forestry proposals but provides little forward visibility on which will advance. Previous work established the descriptive and historical baseline and introduced two probabilistic models, a Bayesian scorecard, a Cox proportional-hazards model, as well as a composite power ranking. This paper documents the deployment of a minimum viable product (MVP): the MPI Dashboard. The dashboard ingests MPI data, applies pre-computed probabilities and urgency measures, and presents blended probabilities, priority indices, and power rankings through five interactive tabs: Probability and Ranking, Sector and Cleantech, Start-Year and Cost, Map, and Stage Flow. Performance testing showed sub-second refresh times and reproducible outputs. The MVP demonstrates that probabilistic triage can be transparent, responsive, and scalable, bridging the upstream intelligence gap and transforming a static registry into a living tool for planning and resource allocation.

Introduction

Canada's major project pipeline now exceeds $1.3 trillion in announced value, yet historical data show that on average one in four proposals move from planning to construction within five years. This gap between ambition and execution risks stranded capital, stakeholder fatigue, and missed climate targets. Existing tools for project oversight, such as Project Portfolio Management (PPM) systems and Governance, Risk, and Compliance (GRC) platforms, typically activate only after a project has passed due diligence or received formal onboarding. This leaves a critical upstream intelligence gap: before an investment decision, sponsors and policymakers lack transparent, repeatable methods to triage large numbers of proposals quickly and defensibly.

Papers 1 and 2 established the descriptive and historical foundation of the MPI, documenting transition pathways, cohort survival, and bottlenecks in Canada's major project pipeline. Papers 3 through 5 then developed the statistical foundations of a forward-looking framework. Paper 3 introduced two lightweight, auditable models: a Bayesian likelihood-ratio scorecard, which estimates the probability that a project will reach construction, and a Cox proportional-hazards model, which estimates the time remaining to that milestone. Paper 4 operationalised these models at the portfolio level, generating blended probabilities and urgency-adjusted priority indices for hundreds of projects in the 2024 MPI. Paper 5 extended this work by introducing power ranking as a composite score that integrates probability and urgency into a single metric for resource allocation across competitive envelopes, demonstrated through a Clean Fuels Fund use case.

This sixth paper advances the framework from research to deployment. It documents the implementation of an MPI Dashboard, a minimum viable product (MVP) platform. The dashboard integrates the Bayes and Cox risk engines, blended probability, priority index, and power ranking into an interactive interface that allows users to filter, rank, and visualize pre-construction projects. By providing real-time, data-driven triage, the MPI Dashboard transforms a static registry into an operational cockpit.

The contribution is twofold. First, it demonstrates how probabilistic risk models can be embedded in a production-grade dashboard that runs quickly, refreshes automatically, and remains fully auditable. Second, it shows how such a system overcomes the upstream intelligence gap, giving developers, planners, financiers, and policymakers a transparent tool for sequencing projects, targeting scarce capital, and aligning policy attention with the projects most likely and most urgent to advance.

Methods

Design basis

The MPI Dashboard was developed as a minimum viable product to demonstrate how probabilistic risk models could be embedded into a transparent, user-facing tool. Its design was guided by four principles.

Integration: The dashboard needed to combine the outputs of the Bayesian likelihood-ratio scorecard and the Cox proportional-hazards model into blended probabilities, urgency measures, and power rankings that could be visualized interactively.

Accessibility: Users required a low-friction interface, accessible through a browser without coding, with filters for province, sector, sponsor, cleantech status, cost band, and year.

Responsiveness: Performance targets were set at under two seconds for initial load and under one second for chart refreshes, even with typical portfolio sizes of 350–400 projects.

Auditability: Outputs had to remain reproducible, with fixed coefficient files, deterministic preprocessing rules, and CSV export for external validation.

System architecture

The system follows a modular pipeline with three layers.

Data ingestion and preprocessing: The dashboard ingests the MPI dataset in Excel format. Required fields include categorical descriptors (province, sector, group, cleantech flag, status codes), project attributes (cost, start and end year, sponsor, project name, coordinates), and model outputs (blended probability, priority index, and power ranking). Input validation enforces schema consistency, coerces types, and preserves unknown values.

Risk engines: Probabilities are pre-computed externally by the Bayesian and Cox models, blended at a 60–40 ratio, and combined with years remaining to yield the priority index. Power ranking integrates blended probability and rescaled priority index into a single score.

Visualization layer: Built in Python using Plotly Dash, the interface exposes the dataset through a web application with dynamic filters, interactive charts, and downloadable outputs. Filters allow users to slice projects by province, sector, sponsor, cleantech status, cost band, and year. KPI cards at the top of the interface display the total number of active projects, the sum of investment value, and the average probability of construction. All views update automatically as filters change.

Results and Discussion

User interface

The interface is organized into five main tabs, each offering a distinct perspective on the dataset.

1. Probability and Ranking: A scatter plot positions projects by probability of construction and urgency (priority index), divided into quadrants by median reference lines. Interactive hovers display company, project, sector, province, cost, and all three metrics (probability, priority, and power ranking). Supplementary bar charts list the Top-N projects by power ranking, by probability, and by priority index, with the number adjustable by a slider.
2. Sector and Cleantech: Stacked bar charts show the distribution of projects by sector and subsector group, broken out by province. A donut chart summarizes the cleantech share of the portfolio.
3. Start-Year and Cost: Time-series bar plots track project initiation by year and sector. Histograms and box plots show the distribution of project costs, highlighting the skew toward a handful of very large projects.
4. Map: A geospatial view places projects on an interactive map using latitude and longitude. Bubble size corresponds to capital cost, and colour distinguishes sector.
5. Stage Flow: A Sankey diagram maps transitions between start and end status codes, providing a flow view of project progression or attrition.

Multi-panel dashboard

Figure 1 presents the five dashboard tabs side by side. Panel (a) shows the Probability and Ranking scatter plot with quadrant overlays and Top-N bar charts. Panel (b) displays sector and cleantech breakdowns. Panel (c) illustrates project counts and costs by start year. Panel (d) shows the geospatial map with proportional bubbles. Panel (e) provides the stage flow diagram. Together these panels demonstrate the dashboard's ability to render multiple complementary views of the project pipeline, all linked by a common filtering logic.

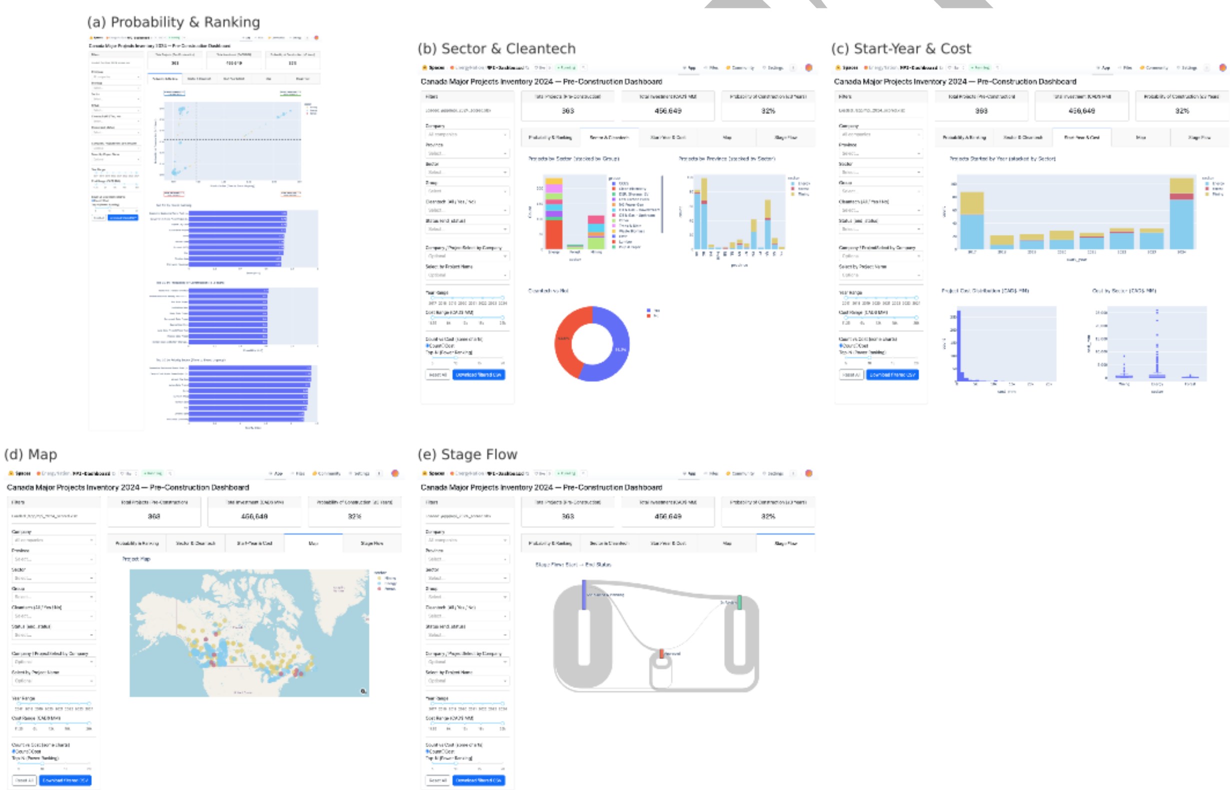

Figure 1: Multi-Panel Dashboard.

Performance and reproducibility

Testing on MPI files of roughly 350–400 projects confirmed that the dashboard met its design basis. Initial load times averaged under two seconds, while filter updates and chart redraws are refreshed in less than one second. CSV export produced identical outputs across sessions, ensuring reproducibility.

User insights

Interactive filtering enabled rapid triage across multiple dimensions. In the Probability and Ranking tab, quadrant overlays separated shovel-ready projects from long-horizon or low-likelihood files. Sector and cleantech charts revealed clustering patterns by region and industry. The geospatial tab highlighted provincial concentrations of high-priority projects, while the stage-flow diagram provided a systemic view of progression and attrition.

Lessons from the MVP

Deployment of the MPI Dashboard demonstrated that probabilistic risk models can be embedded into a responsive, user-facing system with relatively light infrastructure. Three lessons stand out.

Transparency is achievable at scale. By exposing blended probabilities, priority indices, and power rankings directly in the interface, the dashboard allows users to trace how each project's score is derived. This builds confidence among stakeholders in contrast to opaque black-box models.

Responsiveness matters as much as accuracy. Even simple models will not be adopted if users must wait for results. Meeting the sub-second refresh thresholds proved critical: planners and policymakers can experiment with filters in real time, encouraging exploration rather than static reporting.

Data quality remains the bottleneck. The risk engines are deterministic and reproducible, but the MPI dataset contains missing values, inconsistent labels, and limited contextual variables. The dashboard mitigates this by coercing unknown categories and enforcing schema rules, but future adoption will depend on cleaner, timelier data.

Practical constraints

The MVP revealed several constraints. Manual file upload complicates version control and makes it difficult to ensure that all users are working from the same dataset. In addition, rigid input schema requirements mean that project files must strictly match expected column names, which can create friction when integrating new or updated data. Contextual blind spots persist, as permitting milestones, Indigenous benefit agreements, or commercial contracts are not yet captured in the MPI dataset. These constraints restrict the current dashboard to exploratory use rather than operational integration.

Next steps

The path forward has three layers.

1. Data integration: Automating ingestion from official MPI releases would eliminate manual uploads and version drift. Adding contextual signals such as regulatory permits or commercial contracts would improve risk discrimination without overcomplicating the models.
2. Platform development: A lightweight API would allow users to score new projects and benchmark them against the existing dataset. Multi-user access, secure hosting, and user authentication would transition the dashboard from MVP to production-ready service.

3. Analytical extensions: Future builds could incorporate dynamic weighting between probability and urgency, simulate funding allocations across competitive envelopes, or link to external indicators. These enhancements would position the dashboard not just as a monitor but as a decision-support tool for governments, financiers, and developers.

Broader implications

The MPI Dashboard shows how upstream intelligence gaps can be bridged by embedding probabilistic models in transparent, interactive systems. It reduces analytic drag, lowers the cost of project surveillance, and provides a reproducible baseline for triage decisions. While further development is needed before operational deployment, the MVP confirms that a lean, auditable dashboard can transform a static project registry into a living tool for resource allocation and strategic planning.

Conclusion

This paper documented the implementation of the MPI Dashboard as a minimum viable product. It completes a sequence of work that began with Papers 1 and 2, which established the descriptive and historical foundations of the MPI through transition analysis, cohort tracking, and bottleneck identification. Papers 3 through 5 then developed the probabilistic risk framework: Paper 3 introduced the Bayesian scorecard and Cox proportional-hazards model, Paper 4 applied them at the portfolio level to generate blended probabilities and priority indices, and Paper 5 introduced power ranking as a composite score to order projects by both feasibility and urgency. Paper 6 brings these elements together in a functioning, user-facing system.

The dashboard met its design basis for integration, accessibility, responsiveness, and auditability. By combining Bayesian and Cox outputs into blended probabilities, priority indices, and power rankings, and presenting them through an interactive interface, it demonstrates how probabilistic intelligence can be delivered in real time. Performance testing confirmed that load and refresh times meet responsiveness thresholds, while reproducibility safeguards ensure consistent outputs across sessions.

Lessons from the minimum viable product are clear. Transparency is achievable at scale when scores are exposed directly to users. Responsiveness is critical for adoption, as real-time updates encourage exploration and trust. Data quality remains the limiting factor, requiring cleaner and timelier inputs to sustain credibility. The MVP also highlighted practical constraints, including manual uploads, rigid schema requirements, and contextual blind spots not yet captured in the MPI dataset.

Future development will focus on automating ingestion of official MPI releases, adding contextual signals such as permitting and commercial milestones, and enabling secure multi-user access through an API. Analytical extensions could allow dynamic weighting between probability and urgency, simulate funding allocations across competitive envelopes, or link to external indicators. With these enhancements, the dashboard can evolve from an exploratory tool into a decision-support cockpit.

By bridging the upstream intelligence gap that precedes conventional portfolio management and compliance systems, the MPI Dashboard transforms a static registry into a living analytical platform. Taken together, Papers 1 and 2 provided the baseline understanding, Papers 3 through 5 built and refined the risk models and ranking framework, and Paper 6 demonstrates deployment. The series as a whole establishes a transparent, auditable, and scalable framework for understanding Canada's major project pipeline and aligning capital and policy attention with the projects most likely, and most urgent, to advance.